F23-NR-601
FIM III BOOSTER STATION TRANSFORMER
F23-TR-601 NEUTRAL EARTHING CUBICLE

254Ω
25A FOR 10s
30/0.05A
SENSING
ER-15kV
NOTE 9

F23-TR-601
33 / 11kV
5000 kVA
Dyn11
z=7.5%
KNAN

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

POLE MOUNTED
RECLOSER PANEL
NOTE 8

FIM III TSF HV DISTRIBUTION
33 kV POWERLINE
POLE TSF3-2

IS1
REC 1
SG
SG
AUX POWER
RECLOSER CONTROL PANEL

F23-SR-601
FIM III BOOSTER STATION HV/LV SWITCHROOM

E-SHUT
F23SR601-LCS
NOTE 5

NOTE 4

F23-SW-601
FIM III BOOSTER STATION HV SWITCHBOARD

TEST
PQM
SEL735
PR
SE-330AU
EARTH FAULT INDICATION
TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)
UPSTREAM TRIPS

500/1A
2.5VA
Cl 0.2
500/1A
2.5VA
5P20
CLOSE
TRIP
630A
NOTE 2
50 50G 51 51G AFD REF 50N 51N
PR
SEL 751
49 63 71

NOTE 2
NOTE 3
A01 A02 A03 A04 A05 A06
NOTE C.1
11kV 1250A 3-E 25kA FOR 3s
110V AC SUPPLY
20A
63A
2A
630A
NOTE 2
VT
11kV/√3 / 0.11kV/√3 / 0.11kV/√3
15VA CLASS 0.2
20VA CLASS 3P
E160A
400A
50/1A
2.5VA
5P20
50/1A
1.5VA
10P10
150/1A
2.5VA
5P20
NOTE 9
VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT
NOTE C.2
F23PP651-VS
FIM III BOOSTER
PUMP 1 VSD
11/3.3kV 800kW
F23PP652-VS
FIM III BOOSTER
PUMP 2 VSD
11/3.3kV 800kW
M
3~
RTD
800kW
F23-PP-651
FINAL TAILS
PUMP 1
NOTE C.3
F23-PP-652
FINAL TAILS
PUMP 2
MOTE C.3
ACTIVATION OF SWRM E-SHUT
DETECTION OF EARTH FAULT
DETECTION OF ARC FAULT
F23-TR-602
FIM III BOOSTER
11/0.433kV TRANSFORMER
F23-SW-602
FIM III BOOSTER
415V MCC INCOMER
11kV SPARE
FULLY EQUIPPED
REFER DRG 2000-F23-SLD-EL-40011

NOTES

1. THE SWITCHBOARD SHALL BE EXTENDABLE AT ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENT IS CURRENT SUPPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE. CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION.
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON IS HARD WIRED DIRECTLY TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND UPSTREAM 33kV FEEDER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE THE UPSTREAM 33kV FEEDER.
7. UPSTREAM 33kV FEEDER SHALL BE TRIPPED ON THE EARTH AND ARC FAULT DETECTION OF THE DOWNSTREAM INCOMER ACB.
8. 33kV POWER SUPPLY EQUIPMENT BY OTHERS, INCLUDING RECLOSER AND RECLOSER CONTROL PANEL.
9. FOR MECHANICAL KEY INTERLOCK DETAILS REFER TO DRAWING NO. 2000-F23-SLD-EL-40002.
10. REFER TO CABLE SCHEDULE 2000-F23-LST-EL-40003.

LEGEND

- PR — INCOMER / FEEDER PROTECTION RELAY
- PQM — POWER QUALITY METER
- ARC FLASH SENSOR (POINT TYPE) NOTE 2, 8
- CAPACITIVE VOLTAGE INDICATION
- KEY-OPERATED INTERLOCK
- MECHANICAL INTERLOCK DEVICE
- KEY EXCHANGE BOX
- E-SHUT — EMERGENCY SHUTDOWN

AS BUILT

2000-F23-SLD-EL-40001_Rev3

Fim III booster station F23-SW-601 HV switchboard single line diagram